

Testing for Significance (F and T) Transcript

Hello everyone, welcome to lesson 2 video 2. In this lesson, we will study testing for significance in regression. This involves using F-tests. And to test these types of tests, I'll let you know about how to do hypothesis testing, frame hypothesis testing using F-tests.

Hey hi everyone. And how do we estimate that the relationship that we are getting is really significant or to better phrase it, how do we say that the relationship is statistically significant? When I say relationship, because we are studying simple linear regression, we could have one y variable and one x variable. And we are trying to find out what is that relationship, right? Just to give you an idea, let's say that we have the first y variable, sorry, the y variable as sales and the x variable as ads.

Now we are trying to find out how does sales of a company gets influenced by ads, right? In this case, we can easily fit in a chart, so insert, go to charts and you see that this is the scatterplot that we have got, this is called a scatterplot because the data is scattered here and there and on the click of the data, you will be able to see what the data point is, okay. Now if you see, let me go here, if you see all of this data, you can think of there is a simple linear relationship between both the variables, between sales and ads, that's what we are trying to do in linear regression, right? We have the data. Using that data, we are trying to find what is that relationship.

Now, let's add that relationship here. Okay, let's try adding the relationship here. Let's say add trend 9, now you see that this is the trend 9 and let's add the r squared value.

You'll see that the r squared value is 0.9, which means 90% of the variation in sales is being explained by ads. Okay, or there are the variations which are happening in sales are more influenced by ads, that's what it means in this particular case, more ads, more sales, that's what this entire distribution of data on this entire scatterplot tells you. But we might get an equation, okay, we might get a relationship which says that sales is equals to, let's say, 10 times of ads plus 90.

So if you get a relationship like this, think of it like, think of it, if you get a relationship like this, how good you are to make an assumption about 10 is a real right estimate for the parameter beta 1, but because beta 1 is the slope parameter, right, if you remember, so this is y , y is equals to $\beta_1 x + \beta_0$. So the estimate for beta 1 is 10 here, how good or how sure you are or how certain you are that the estimate of beta 1 that you have got here, which is 10 in this case, we are assuming this though, but how certain that this 10 value is the most realistic value or most statistically significant value, okay, that's what we are going to learn in this entire session. I hope this will give you an understanding about how do we get to know whether the sample estimate of the slope parameter is statistically significant or not. So the very first thing is why we are thinking of samples here, we are thinking of samples because we generally do not get the entire population data, we'll be only having a sample and using that sample, we'll be doing the study on the entire population.

So in this case, if you think of this particular case, you have a sample of maybe last one years of data, right? And then using that you'll you're trying to say, how does sales in general is dependent on ads, right? You have a sample, using that sample, you are trying to generalise the relationship between sales and ads. So you are trying to generalise it to the wider population in turn. So that's why that sample comes into picture.

And because we are talking about sample, we are not talking about any other parameter, we are talking about sample estimate of our slope parameter. And why are we saying whether slope is zero or not? Think of the equation here. If you think of this is the current estimate, right, 10 is the current estimate of the slope parameter.

What if this becomes zero? If it becomes zero, it means that sales is not at all dependent on ads. If you think of that way, right, if I zoom it up now, if 10 becomes zero, it means sales is not at all dependent on ads. And that's what is important for us to know, whether the parameter, beta 1 parameter, the slope parameter is actually zero or it is not zero, right? The key question is our sample slope, beta 1, how far is it from zero? Is it actually zero or it is non-zero? For doing that, we do t-tests.

For t-test, we first hypothesise about the situation. We say that our null hypothesis is there is no relationship between Y and X. Our alternate hypothesis is there is relationship between Y and X, which in that case, beta 1 would be not zero. It can be positive, negative, but basically non-zero.

Then we compute t-statistics as a difference between what you have got as a sample estimate of beta 1. Here in this case, let's say the data, the sample estimate is 10 and 90. 90 is the intercept estimate, 10 is the slope estimate. We are testing for slope.

OK, so herein, you'll be getting 10 minus zero because this is what we have hypothesised in our null hypothesis that in general it should be zero. But if there are chances, those chances are very rare, then we'll say that the beta 1 parameter is actually not zero, right? In this particular case, we have beta 1 value, which we have deduced on the sample. We see how far that is from the hypothesised value, right? Divided by the standard error of that.

Now, if T value is significant, then in that case, P value would be less than 0.05, given that the level of significance is 5%. If P value is less than 0.05, it means that the alternate hypothesis is true, which means beta 1 value that you have got or non-zero beta 1 value that you have got on sample is actually non-zero in the population. Now, to test for the entire model, we use F-statistics as well.

Now, F-statistics become more relevant in multiple linear regression because F-statistics and simple linear regression is nothing but the square root of T-statistic, OK? When you have simple linear regression, the idea is to explain how much amount of variance in Y variable is coming from the X variable. And how much amount of that is coming from the error part, right? Remember, you assume like $Y = \beta_0 + \beta_1 X + E$ in previous lectures. Hence, you are trying to find out

how much amount of variation in Y is composed of variation due to the factors that you have included or the X variable and variation due to error, OK? Now, if this statistic is greater than 1, which eventually means that the numerator is greater than the denominator, in that case, there are more chances.

I'm not saying that you're sure. In that case, there are more chances that your model of your X variable is actually contributing to the change in Y, OK? So, the variation in Y is being contributed by X variable. That's what it means.

We'll look into it in more details in multiple linear regression. But just to help you understand, F statistics is almost similar to what your T statistics is. In most of the packages, you will be getting an output like this.

Whether you use JMP, Minitab, R, Python or whatever, you'll get an output like this. Herein, you will be getting an output like what is the value of R square, which is nothing but amount of variance in Y explained by X variable. Then you will be getting an ANOVA table, which is for F statistic.

Now, if F statistics is significant, which means that F greater than 1, it means that the numerator in the previous slide, right? If you remember the variance explained by the model, it means that model is explaining the variance more. Hence, P value is less than, sorry, less than 0.05. And hence, you say that the model is significant, right? So F statistics is used for saying whether the model that you're making is significant or not. Then you will go into individual terms.

The very first is for intercept term and the second line is the slope term. The second line is more important, which I have highlighted because if the coefficient of ad spend is 0, then there is no relationship. So you forget about the linear regression in that case, right? So if ad spend is not 0, you are trying to understand because this is what you have got from the sample.

You are trying to understand how significant this value is. Should we rely on this or should we not, right? So then you do T statistics, T test. If in your T test, you get a significant value, which can be depicted by P value.

If P value is less than 0.05, it means that the value that you have got is actually significant. It means the sales is actually dependent on ad spend. In summary, the F test tells us if the regression model as a whole is a significant while T test tells us whether the coefficients are significant.

Together, they confirm whether the regression is a valid tool or it is not right to explain the relationship between independent and dependent variables.

-----End-----